=== UNIT 4: QUANTUM MECHANICS & WAVE-PARTICLE DUALITY ===

>> 4.1 Inadequacy of Classical Physics and the Quantum Hypothesis

Classical Newtonian mechanics and Maxwellian electrodynamics fail to explain microscopic subatomic phenomena:
* Ultraviolet Catastrophe: Classical Rayleigh-Jeans blackbody law predicted radiation intensity diverges to infinity at high frequencies (u(nu) proportional to nu^2). Resolved by Max Planck (1900) by hypothesizing that energy exchange occurs in discrete quantum packets: E = n * h * nu, where h = 6.626 * 10^-34 J-s.
* Photoelectric Effect: Emission of electrons from metals illuminated by light. Heinrich Hertz / Albert Einstein (1905) demonstrated that: (1) Emission occurs instantaneously without time lag, (2) Kinetic energy of photoelectrons depends strictly on photon frequency, NOT light intensity: KE_max = h*nu - Phi_0 (where Phi_0 is work function), (3) Threshold frequency nu_0 below which no emission occurs regardless of intensity.
* Compton Scattering: Inelastic scattering of X-ray photons by stationary electrons resulting in wavelength shift: Delta lambda = lambda' - lambda = (h / (m_e * c)) * (1 - cos(theta)), proving that photons carry discrete relativistic momentum p = h / lambda.

>> 4.2 De Broglie Hypothesis of Matter Waves

In 1924, Louis de Broglie hypothesized that if light exhibits wave-particle duality, matter (electrons, protons, atoms) must likewise exhibit wave properties:
* De Broglie Wavelength: lambda = h / p = h / (m * v).
* Non-Relativistic Kinetic Energy Relation: p = sqrt(2 * m * E) => lambda = h / sqrt(2 * m * E).
* For an electron accelerated from rest across potential difference V: E = e * V => lambda = h / sqrt(2 * m_e * e * V) = 1.227 / sqrt(V) nanometers.
* Davisson-Germer Experiment (1927): Directed a collimated beam of 54 V electrons onto a nickel target and observed intense constructive diffraction peaks at scattering angle theta = 50 deg, exactly confirming Bragg's wave law and proving matter waves experimentally.

>> 4.3 Heisenberg's Uncertainty Principle

Werner Heisenberg (1927) proved that conjugate dynamical variables cannot be simultaneously measured with arbitrary precision:
* Position-Momentum Uncertainty: Delta x * Delta p_x >= h_bar / 2, where h_bar = h / (2 * pi) = 1.05457 * 10^-34 J-s.
* Energy-Time Uncertainty: Delta E * Delta t >= h_bar / 2. Explains natural spectral

line broadening of atomic transitions with finite lifetime.
* Proof that Electrons Cannot Reside in Atomic Nucleus: Nuclear radius R_n ~ 10^-14 m. If an electron were confined inside: Delta x ~ 10^-14 m => Delta p >= h_bar / (2 * Delta x) ~ 5.3 * 10^-21 kg-m/s. Kinetic energy would exceed 20 MeV. However, beta decay electrons have energies < 3 MeV, proving electrons cannot exist inside nuclei.

>> 4.4 The Wave Function and Born's Statistical Interpretation

* Wave Function Psi(x,t): A complex quantity containing all dynamical information about a quantum system: Psi(x,t) = psi(x) * e^(- i * E * t / h_bar).
* Born's Probability Interpretation: Max Born (1926) interpreted |Psi(x,t)|^2 = Psi * Psi^* as Probability Density: the probability dP of finding the particle in volume element dx is dP = |Psi(x,t)|^2 dx.
* Physical Well-Behaved Conditions on Wave Function: (1) Must be continuous everywhere, (2) First spatial derivative dPsi/dx must be continuous (except at infinite potential barriers), (3) Single-valued everywhere, (4) Square-integrable (Normalizable).
* Normalization Condition: Total probability of finding the particle across all space must equal unity: Integral_-inf^+inf |Psi(x,t)|^2 dx = 1.

>> 4.5 Quantum Mechanical Operators and Schrodinger Equations

In quantum mechanics, physical observables are represented by linear Hermitian operators acting on wave functions:
* Position Operator: x_hat = x.
* Linear Momentum Operator: p_hat = - i * h_bar * (d/dx). In 3D: p_hat = - i * h_bar * Del.
* Kinetic Energy Operator: T_hat = p_hat^2 / (2*m) = - (h_bar^2 / 2m) * (d^2/dx^2).
* Total Energy (Hamiltonian) Operator: H_hat = T_hat + V_hat = - (h_bar^2 / 2m) * (d^2/dx^2) + V(x).
* Time-Dependent Schrodinger Equation (TDSE): i * h_bar * (dPsi/dt) = H_hat * Psi = [- (h_bar^2 / 2m) * Del^2 + V(r)] * Psi.
* Time-Independent Schrodinger Equation (TISE): Separation of variables yields: [- (h_bar^2 / 2m) * (d^2psi/dx^2) + V(x) * psi(x)] = E * psi(x).

>> 4.6 Particle in a One-Dimensional Infinite Potential Box

Consider a particle of mass m confined in an infinitely deep 1D potential well: V(x) = 0 for 0 < x < L, and V(x) = infinity for x <= 0 and x >= L:
* Boundary Conditions: psi(0) = 0, and psi(L) = 0 (particle cannot penetrate infinite walls).
* Eigen Wave Functions: psi_n(x) = sqrt(2 / L) * sin(n * pi * x / L), where n = 1, 2, 3, ... (Quantum Number).

* Quantized Energy Eigenvalues: E_n = (n^2 * pi^2 * h_bar^2) / (2 * m * L^2) = (n^2 * h^2) / (8 * m * L^2).
* Zero-Point Energy: Ground state (n = 1) energy E1 = h^2 / (8 * m * L^2) > 0. A particle in a box can NEVER have zero kinetic energy (consistent with Heisenberg uncertainty principle: if E=0, p=0 exactly => Delta x = infinity, violating box confinement).

[EXAMPLE 1: De Broglie Wavelength: Electron vs Macroscopic Object]

Problem Context: Compare the wave properties of two entities: (1) An electron accelerated through potential difference V = 100 Volts (mass m_e = 9.109 * 10^-31 kg), (2) A macroscopic golf ball of mass m = 46 g (0.046 kg) flying at velocity v = 30 m/s. Explain why wave mechanics governs the electron while classical mechanics suffices for the golf ball.
Solution Step 1: De Broglie wavelength of accelerated electron:
lambda_e = 1.227 / sqrt(V) nm = 1.227 / sqrt(100) = 1.227 / 10 = 0.1227 nm = 1.227 * 10^-10 m (1.227 Angstroms).
Physical significance: 0.123 nm is of the exact same order of magnitude as atomic interplanar spacings in solids (d ~ 0.1 to 0.3 nm), resulting in measurable wave diffraction.
Solution Step 2: De Broglie wavelength of macroscopic ball:
Momentum p = m * v = 0.046 kg * 30 m/s = 1.38 kg-m/s.
lambda_ball = h / p = (6.626 * 10^-34 J-s) / 1.38 kg-m/s = 4.80 * 10^-34 m.
Physical significance: 10^-34 m is 10^20 times smaller than the diameter of a proton (~10^-15 m)! No physical aperture or crystal lattice can diffract such wavelengths.
Conclusion: Electron wavelength (0.123 nm) produces noticeable quantum wave phenomena, whereas macroscopic wavelength (10^-34 m) is completely imperceptible.

[EXAMPLE 2: Proof that Electron Cannot Exist in Atomic Nucleus]

Problem Context: The typical radius of an atomic nucleus is R = 1.5 * 10^-15 m (1.5 femtometers). Assuming an electron were confined inside, use Heisenberg's Uncertainty Principle to compute: (1) Minimum uncertainty in momentum, (2) Minimum kinetic energy, (3) Compare with observed beta-particle energies.
Solution Step 1: Position Uncertainty Delta x:
Maximum diameter of confinement: Delta x = 2 * R = 3.0 * 10^-15 m.
Solution Step 2: Momentum Uncertainty Delta p:
Delta x * Delta p >= h_bar / 2 => Delta p >= h_bar / (2 * Delta x).
Delta p >= (1.05457 * 10^-34 J-s) / (2 * 3.0 * 10^-15 m) = 1.05457 * 10^-34 / 6.0 * 10^-15 = 1.758 * 10^-20 kg-m/s.
Hence minimum momentum magnitude p ~= Delta p = 1.758 * 10^-20 kg-m/s.
Solution Step 3: Relativistic Kinetic Energy:

Since p * c = (1.758 * 10^-20) * (3.0 * 10^8) = 5.274 * 10^-12 J >> m_e * c^2 (8.19 * 10^-14 J), motion is ultra-relativistic:
E_total ~= p * c = 5.274 * 10^-12 J / (1.602 * 10^-13 J/MeV) = 32.92 MeV.
Minimum KE ~= E_total - m_e*c^2 ~= 32.92 - 0.511 = 32.41 MeV.
Solution Step 4: Physical Deduction:
Experimental beta decay emissions from radioactive nuclei yield electrons with kinetic energies of at most 2 to 3 MeV, NEVER 32 MeV.
Conclusion: An electron cannot reside as an intrinsic structural constituent inside an atomic nucleus; beta decay electrons are synthesized at the instant of decay via neutron-proton conversion.

[EXAMPLE 3: Particle in 1D Well: Normalization & Finding Probability]

Problem Context: A quantum particle of mass m is trapped in an infinite potential box of width L (0 <= x <= L) in its first excited state (n = 2), described by wave function psi_2(x) = A * sin(2*pi*x / L). (1) Find normalization constant A, (2) Calculate probability of finding particle in central region between x = L/4 and x = 3*L/4.
Solution Step 1: Normalization Integral:
Integral_0^L |psi_2(x)|^2 dx = 1 => A^2 * Integral_0^L sin^2(2*pi*x / L) dx = 1.
Using identity sin^2(theta) = (1 - cos(2*theta)) / 2:
A^2 * Integral_0^L [0.5 - 0.5 * cos(4*pi*x / L)] dx = A^2 * [0.5*x - (L / 8*pi)*sin(4*pi*x / L)]_0^L.
A^2 * [0.5*L - 0] = 1 => A^2 * (L / 2) = 1 => A = sqrt(2 / L).
Solution Step 2: Probability in Central Half (L/4 <= x <= 3*L/4):
P = Integral_{L/4}^{3L/4} (2 / L) * sin^2(2*pi*x / L) dx = (2 / L) * [x/2 - (L / 8*pi)*sin(4*pi*x / L)]_{L/4}^{3L/4}.
Evaluate at 3*L/4: sin(4*pi*(3L/4)/L) = sin(3*pi) = 0.
Evaluate at L/4: sin(4*pi*(L/4)/L) = sin(pi) = 0.
P = (2 / L) * [(3*L/8 - 0) - (L/8 - 0)] = (2 / L) * (2*L / 8) = (2 / L) * (L / 4) = 2 / 4 = 0.50 (50.0%).
Conclusion: Normalization constant A = sqrt(2/L); Probability of finding particle in central region is exactly 50%.

[EXAMPLE 4: Energy Levels of Electron in a 1 nm Quantum Well]

Problem Context: An electron (m_e = 9.109 * 10^-31 kg) is confined in a 1D semiconductor quantum well of width L = 1.0 nm (1.0 * 10^-9 m). Calculate: (1) Ground-state energy E1 in eV, (2) Energies of states n = 2 and n = 3, (3) Wavelength of photon emitted during transition from n = 3 to n = 1.
Solution Step 1: Ground-State Energy E1:
E1 = h^2 / (8 * m_e * L^2) = (6.626 * 10^-34)^2 / [8 * 9.109 * 10^-31 * (1.0 * 10^-9)^2].

E1 = (4.3904 * 10^-67) / [7.2872 * 10^-48] = 6.0248 * 10^-20 Joules.
In eV: E1 = 6.0248 * 10^-20 J / (1.6022 * 10^-19 J/eV) = 0.3760 eV.
Solution Step 2: Energy of Excited States E2 and E3 (En = n^2 * E1):
E2 = 2^2 * E1 = 4 * 0.3760 eV = 1.504 eV.
E3 = 3^2 * E1 = 9 * 0.3760 eV = 3.384 eV.
Solution Step 3: Transition Wavelength lambda (n = 3 to n = 1):
Delta E = E3 - E1 = 3.384 - 0.376 = 3.008 eV = 4.819 * 10^-19 Joules.
lambda = (h * c) / Delta E = (6.626 * 10^-34 * 3.0 * 10^8) / (4.819 * 10^-19) = (1.9878 * 10^-25) / 4.819 * 10^-19 = 4.125 * 10^-7 m = 412.5 nm (violet light).
Conclusion: E1 = 0.376 eV; E2 = 1.504 eV; E3 = 3.384 eV; Emitted photon wavelength = 412.5 nm.

>> 4.7 University Review & Examination Questions

* Q1: State the fundamental postulates of quantum mechanics and derive the time-independent Schrodinger wave equation.
Answer: Postulates: (1) State of a quantum particle is completely specified by wave function Psi(r,t). (2) Physical observables correspond to linear Hermitian operators. (3) The only possible measured values of observable A are eigenvalues a_n of equation A_hat*psi_n = a_n*psi_n. (4) Probability density is |Psi|^2. Derivation of TISE: Total energy equation is E = T + V = p^2/(2m) + V. Replace dynamical variables with operators: E -> i*h_bar*(d/dt), p -> -i*h_bar*(d/dx). Operating on Psi(x,t): i*h_bar*(dPsi/dt) = -(h_bar^2 / 2m) * (d^2Psi/dx^2) + V(x)*Psi. For stationary states, separate variables: Psi(x,t) = psi(x) * e^(- i*E*t / h_bar). Substituting and dividing by e^(- i*E*t / h_bar) yields: - (h_bar^2 / 2m) * (d^2psi/dx^2) + V(x)*psi(x) = E*psi(x).
* Q2: Solve the Schrodinger equation for a particle in a 1D infinitely deep potential well and discuss the physical meaning of zero-point energy.
Answer: Inside the well (0 < x < L), V = 0, so d^2psi/dx^2 + k^2*psi = 0, where k = sqrt(2mE)/h_bar. General solution is psi(x) = A*sin(kx) + B*cos(kx). Boundary condition psi(0) = 0 requires B = 0. Boundary condition psi(L) = 0 requires A*sin(kL) = 0 => kL = n*pi (n = 1,2,3...). Hence k = n*pi/L. Energy eigenvalues are E_n = (h_bar^2*k^2)/(2m) = (n^2*pi^2*h_bar^2)/(2mL^2) = (n^2*h^2)/(8mL^2). Normalization gives A = sqrt(2/L), so psi_n(x) = sqrt(2/L)*sin(n*pi*x/L). Zero-Point Energy: The lowest state n = 1 has energy E1 = h^2/(8mL^2) > 0. A particle in a box cannot have zero energy because by Heisenberg's principle, if E=0, p=0 (Delta p = 0), implying infinite position uncertainty Delta x = infinity, which contradicts spatial confinement within width L.

>> 4.8 Key Takeaways and Revision Points

* De Broglie wavelength lambda = h/p connects wave and particle domains.
* Heisenberg principle Delta x * Delta p >= h_bar/2 forbids simultaneous exact measurement.
* Born interpretation: |Psi|^2 dx represents the probability of finding the particle in interval dx.

* Particle in a box has quantized energies E_n proportional to n^2 and non-zero ground state energy.